

```
In [15]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
In [16]: df=pd.read_csv('Iris.csv')
df
```

```
Out[16]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
In [17]: df.nunique()
```

```
Out[17]: Id          150
SepalLengthCm      35
SepalWidthCm       23
PetalLengthCm      43
PetalWidthCm       22
Species            3
dtype: int64
```

```
In [25]: X = df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
Y = df['Species']
```

```
In [26]: X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_sta
```

```
In [27]: dtree = DecisionTreeClassifier(criterion="gini",max_depth=3, random_state=42)
dtree.fit(X_train , Y_train)
```

```
Out[27]: ▼ DecisionTreeClassifier ⓘ ?  
         ► Parameters
```

```
In [28]: Y_pred = dtree.predict(X_test)
```

```
In [29]: print("Accuracy:", accuracy_score(Y_test, Y_pred))  
         print("\nConfusion Matrix:\n", confusion_matrix(Y_test, Y_pred))
```

Accuracy: 1.0

Confusion Matrix:

```
[[10  0  0]  
 [ 0  9  0]  
 [ 0  0 11]]
```

```
In [30]: import matplotlib.pyplot as plt  
         from sklearn.tree import plot_tree  
  
         plt.figure(figsize=(15,10))  
  
         plot_tree(dtree,  
                   feature_names=X.columns,  
                   class_names=dtree.classes_,  
                   filled=True,  
                   rounded=True  
         )  
         plt.show()
```



